

Environmental Toxicology

Acute and Chronic Toxicity of Cobalt to Freshwater Organisms: Using a Species Sensitivity Distribution Approach to Establish International Water Quality Standards

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Abstract: Water quality standards for cobalt (Co) have not been developed for the European Union or United States. The objective of the present study was to produce freshwater Co toxicity data that could be used by both the European Union and the United States to develop appropriate regulatory standards (i.e., environmental quality standards or predicted-no-effect concentrations in Europe and ambient water quality criteria or state water quality standards in the United States). Eleven species, including algae, an aquatic plant, and several invertebrate and fish species, were used in the performance of acute and chronic Co toxicity tests. Acute median lethal or median effective concentration (LC50 or EC50) values ranged from 90.1 µg Co/L for duckweed (*Lemna minor*) to 157 000 µg Co/L for midges (*Chironomus tentans*). Chronic 10% effect concentration (EC10) values ranged from 4.9 µg Co/L for duckweed to 2170 µg Co/L for rainbow trout (*Oncorhynchus mykiss*). Chronic 20% effect concentration (EC20) values ranged from 11.1 µg Co/L for water flea (*Ceriodaphnia dubia*) to 2495 µg Co/L for *O. mykiss*. Results indicated that invertebrate and algae/plant species are more sensitive to chronic Co exposures than fish. Acute-to-chronic ratios (derived as acute LC50s divided by chronic EC20s) were lowest for juvenile *O. mykiss* (0.6) and highest for the snail *Lymnaea stagnalis* (2670). Following the European-based approach and using EC10 values, species sensitivity distributions (SSDs) were developed and a median hazardous concentration for 5% of the organisms of 1.80 µg Co/L was derived. Chronic EC20 values were used, also in an SSD approach, to derive a US Environmental Protection Agency-style final chronic value of 7.13 µg Co/L. *Environ Toxicol Chem* 2020;39:799–811. © 2020 SETAC

Keywords: Cobalt; Predicted-no-effect concentration; Water quality criteria; Water quality standards; Water quality guidelines

INTRODUCTION

Cobalt (Co) is a naturally occurring element that is ubiquitously distributed throughout the world. It is an anthropogenically important trace metal that is used in a wide range of industrial and technological applications, including use as a component in high temperature-resistant alloys for jet engines and a component that improves the performance of magnetic alloys and rechargeable batteries (Hamilton 1994). It is also used as a pigment, a petroleum industry catalyst, and a nutrient in fertilizers. Cobalt is primarily found in nickel and

nickel–copper deposits and mined in Australia, Canada, Russia, the Democratic Republic of the Congo, and Zambia (Slack et al. 2017; British Geological Survey 2018). Cobalt is usually rare in freshwater environments, other than in situations where mining, manufacturing, or other activities may produce locally elevated concentrations. Gaillardet et al. (2007) estimated an average dissolved Co concentration in major world rivers of 0.148 µg/L, with average values for large rivers around the world ranging from <0.02 to 0.43 µg/L. In Canada, nationwide water quality monitoring showed that dissolved and total Co concentrations ranged from 0.002 to 64 µg/L and from 0.002 to 3.9 µg/L, respectively (Environment Canada 2017). Concentrations reported in surface waters taken from 3 of the most highly contaminated sites in Canada ranged from 0.025 to 2028 µg/L (Environment Canada 2017). In the United States, Co concentrations in surface waters ranged from <0.1 to >1000 µg/L,

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where the highest values were from small streams contaminated by mining (Mebane et al. 2015). Background levels of dissolved Co in European waters are estimated to be 0.18 to 0.21 µg/L (Borg 1983; Neal et al. 1996). Recently, the Forum of European Geological Surveys (Salminen et al. 2005) sampled stream waters from random locations within a grid throughout Europe and determined concentrations of an extensive range of elements, including Co. Mean stream concentrations of dissolved (<0.45 µm) Co were determined to be 0.333 ± 1.01 µg Co/L ($n = 807$, range, 0.01–15.7 µg Co/L).

Although Co is an essential element found in vitamin B₁₂, only trace amounts are required for life (Eitinger et al. 2005; Gruber et al. 2011; Blust 2012). It is unclear what environmental concentrations pose a risk to aquatic species; ambient water quality criteria (AWQC; US Environmental Protection Agency [USEPA]) and environmental quality standards (EQS; European Union under the European Water Framework Directive) do not exist for Co. Canada, however, published a Federal Water Quality Guideline (FWQG) in 2017 that reflected the effect of water hardness (as milligrams of CaCO₃ per liter) on Co bio-availability (Environment Canada 2017); the site-specific value (as dissolved micrograms of Co per liter) is calculated from the following equation:

$$\text{FWQG} = 10^{(0.414[\ln \text{ hardness}] - 1.887)}$$

Few published studies have addressed the toxicity of Co to freshwater organisms. These studies are limited by the number of species tested (i.e., single species; *Daphnia magna* [Biesinger and Christensen 1972], *Oncorhynchus mykiss* [Marr et al. 1998], *Chlamydomonas reinhardtii* [Lustigman et al. 1995]) and the exposure duration, investigating only acute or chronic endpoints. Furthermore, few of the extant studies would satisfy current acceptability requirements for derivation of AWQC or EQS/predicted-no-effect concentration (PNEC) because of shortcomings in the experimental design (e.g., tests not

performed according to standard guidelines, water physicochemical parameters not reported, and results based on nominal concentrations). Although the combination of existing studies provides a useful indication of Co toxicity to aquatic species, the data are too limited or unsuitable to meet regulatory standards. The paucity of Co freshwater toxicity data warrants a comprehensive analysis to determine safe environmental Co concentrations. Such an analysis should take into consideration international data requirements to be useful to all interested global regulatory authorities (e.g., USEPA and the European Chemicals Agency [ECHA]), which may have different toxicity testing criteria and requirements. For example, although the USEPA accepts acute toxicity data for developing AWQC, the ECHA focuses interest predominantly on chronic toxicity data. In addition, the ECHA requires data for algae and higher plants and considers these results when developing a species sensitivity distribution (SSD; Table 1), whereas the USEPA only considers algae/plant data when developing an AWQC (criteria continuous concentration [CCC]) but not as part of an SSD analysis.

The primary objective of this research was to generate a high-quality freshwater toxicity data set for Co that can be used to derive both EQS/PNEC and AWQC. To achieve this, we adopted a tiered approach focused on 1) generation of acute toxicity data needed to derive a USEPA acute AWQC (criteria maximum concentration [CMC]) and set exposure concentrations for chronic tests, and 2) generation of chronic toxicity data needed to derive an European Union EQS/PNEC and USEPA chronic water quality criteria (CCC).

MATERIALS AND METHODS

The research presented herein was conducted as a cooperative effort between the Oregon State University's Aquatic Toxicology Research Laboratory (Albany, OR, USA; formerly Parametrix Environmental Research Laboratory) and the

TABLE 1: Minimum taxonomic group requirements for the European Union species sensitivity distribution method (European Chemicals Agency 2008) and the US Environmental Protection Agency (1985) method for deriving predicted-no-effect concentration or water quality criteria and species and test methods used in the present study to fulfill these requirements

EU requirement	USEPA requirement	Test species	Test method
Fish	The family Salmonidae in the class Osteichthyes	<i>Oncorhynchus mykiss</i>	Early life stage
Second family in the phylum Chordata	A second family of fish in the class Osteichthyes (preferably a commercially or recreationally important warm-water species)	<i>Pimephales promelas</i>	Early life stage
Crustacean	A third family in the phylum Chordata	<i>Danio rerio</i>	Early life stage
Insect	Planktonic crustacean	<i>Daphnia magna</i> , <i>Ceriodaphnia dubia</i>	Life cycle
A family in a phylum other than Arthropoda or Chordata	Insect	<i>Chironomus tentans</i>	Life cycle
	A family in a phylum other than Arthropoda or Chordata	<i>Aeolosoma</i> sp. (phylum Annelida), <i>Brachionus calyciflorus</i> (phylum Rotifera)	Life cycle
A family in any order of insect or any phylum not already represented	A family in any order of insect or any phylum not already represented	<i>Lymnaea stagnalis</i> (phylum Mollusca)	Life cycle
Algae	Benthic crustacean	<i>Hyalella azteca</i>	Chronic, growth
Higher plant		<i>Pseudokirchneriella subcapitata</i>	Chronic, growth
		<i>Lemna minor</i>	Chronic, growth

EU = European Union; USEPA = US Environmental Protection Agency.

University of Ghent's Laboratory of Environmental Toxicology and Aquatic Ecology (Ghent, Belgium).

Chemicals and materials

Oregon State University studies. Reagent-grade cobalt chloride ($\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$; Chemical Abstracts Service no. 7791-13-1; supplied by J.T. Baker, Alfa Aesar, or an equivalent supplier) was used as the test material for all studies. For each study, a Co stock solution was prepared in deionized water and diluted to the exposure concentration with either reconstituted laboratory water, standard synthetic freshwater, or laboratory-blended water (well water blended with reverse osmosis-treated water with a targeted hardness of 80–120 mg/L as CaCO_3). For studies that were conducted using flow-through test procedures (acute and chronic studies with *Pimephales promelas*, *Danio rerio*, and *O. mykiss*), a continuous-flow proportional diluter, designed after Benoit et al. (1982), was employed. The diluter system was constructed of glass, silicone adhesive, and silicone stoppers. Test solutions were delivered to the test chambers through cross-linked polyethylene tubing. A flow-splitting chamber was used for each test concentration to promote mixing of the test solution and to equally allocate the test solution between replicate test chambers. A Marriotte bottle was used to continuously deliver an appropriate volume of Co stock solution to the diluter.

University of Ghent studies. Cobalt chloride ($\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$) was the substance used for all studies. *Daphnia magna* was exposed to $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ in standardized test medium (International Organization for Standardization medium; Organisation for Economic Co-operation and Development [OECD] guideline 211 [Organisation for Economic Co-operation and Development 1998]), which was vigorously aerated prior to spiking. Stock solutions were prepared by dissolving 79.70 or 19.93 mg/L $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ in 100 mL deionized water for the acute and chronic tests, respectively. The entire amount of stock for the tests was prepared at one time: 200 mL (acute) or 5 L (chronic). Exposure solutions for *Hyalella azteca* acute and chronic tests were mixed in a standardized artificial test medium (Borgmann medium [Borgmann 1996]), and stocks were prepared as described for *D. magna*. *Lymnaea stagnalis* tests were conducted using general ASTM International guidelines modified after Gomot (1998), which used artificial freshwater (hardness 140 mg/L as CaCO_3). Test medium for *Pseudokirchneriella subcapitata* was prepared as recommended by OECD guideline 201 (Organisation for Economic Co-operation and Development 1984a) but with ethylenediaminetetraacetic acid (EDTA) replaced by dissolved organic matter (DOM) composed of humic acid at 32 $\mu\text{g/L}$ (Sigma-Aldrich; Heijerick et al. 2002). Similarly, *Lemna minor* was tested in Swedish standard *Lemna* growth medium with EDTA replaced with artificial DOM composed of humic acid at 32 $\mu\text{g/L}$.

Toxicity tests

A series of acute and chronic toxicity tests were conducted to determine the freshwater toxicity of Co to 10 standard

freshwater species. The Oregon State University laboratory conducted tests on *Ceriodaphnia dubia* (water flea), *Aelosoma* sp. (oligochaete), *P. promelas* (fathead minnow), *D. rerio* (zebrafish), and *O. mykiss* (rainbow trout); and the University of Ghent conducted tests on *D. magna* (cladoceran), *H. azteca* (amphipod), *L. stagnalis* (great pond snail), *L. minor* (duckweed), and *P. subcapitata* (green alga). Although 9 aquatic species were subjected to both acute and chronic toxicity tests, *L. minor* and *P. subcapitata* were only used in chronic toxicity testing; however, acute (50% effect concentration [EC50]) and chronic (EC10) values were determined from these tests as described (European Chemicals Agency 2008). Acute and chronic toxicity data for an additional, twelfth species, *Chironomus tentans*, was obtained from a study conducted and reported by Pacific EcoRisk (2005). The acute tests (Table 2) examining *C. tentans* were initiated with second- to third-instar larvae, whereas the chronic tests (Table 3) were initiated with first-instar larvae from an in-house culture. Chronic tests of *C. tentans* were based on the USEPA freshwater sediment chronic toxicity testing guidelines (US Environmental Protection Agency 2000), modified to incorporate use of very fine-grained silica sand. The Co toxicant stock was prepared with CoSO_4 ("Baker Analyzed"; VWR International) rather than CoCl_2 . Exposures were conducted in a flow-through system using both a natural water (Panther Creek, ID, USA; approximate water quality, hardness 35 mg/L as CaCO_3 , pH 8.0, DOC 2.3 mg/L) and a laboratory water as controls.

The acute and chronic toxicity tests conducted at Oregon State University and the University of Ghent laboratories followed the general methodological guidance provided by ASTM International (2002a, 2002b), the US Environmental Protection Agency (2002a, 2002b) and the Organization for Economic Co-operation and Development (1992), unless a standardized protocol was not available. Each acute study was designed to yield median lethal concentration (LC50) and EC50 values following continuous exposure. Details regarding lighting, temperature, species, and life stages used at test initiation; number of replicates; exposure duration; type of exposure (i.e., static, flow-through, etc.); and control/dilution water constitution for acute toxicity tests are summarized in Table 2.

Chronic tests were designed to derive effective concentrations to reduce endpoint performance by 10% and 20% (EC10 and EC20), relative to control. Details regarding lighting, temperature, species, and life stages used at test initiation; number of replicates; exposure duration; type of exposure (i.e., static, flow-through, etc.); and control/dilution water constitution for chronic toxicity tests are summarized in Tables 3 and 4. The chronic tests were conducted using the same species tested in the acute studies, in addition to *P. subcapitata* and *L. minor* (Table 3). The ages at initiation of the chronic tests were the same as those used for acute tests for all species, with the exception that the fish chronic tests were initiated using freshly fertilized eggs (Table 4).

Control and dilution waters were run concurrently with all laboratory tests and comprised of synthetic water prepared by adding salts (e.g., CaSO_4 , MgCl_2) to deionized water. Conductivity (microsiemens per centimeter), hardness

TABLE 2: Overview of the acute test conditions with cobalt chloride for 9 different test species tested at Parametrix/Oregon State University and Ghent University (*Chironomus tentans* tests were conducted in an independent study by Pacific EcoRisk)

	Cladoceran (<i>Daphnia magna</i>)	Amphipod (<i>Hyalella azteca</i>)	Great pond snail (<i>Lymnaea stagnalis</i>)	Water flea (<i>Ceriodaphnia dubia</i>)	Midge (<i>Chironomus tentans</i>)	Oligochaete (<i>Aeolosoma</i> sp.)	Fathead minnow (<i>Pimephales promelas</i>)	Rainbow trout (<i>Oncorhynchus mykiss</i>)	Zebrafish (<i>Danio rerio</i>)	
Origin	Originally collected clone (K6) from a pond in Kiel (Antwerp, Belgium) and cultured for 15 + yr at Ghent University (Belgium)	Aqua Survey (USA) culture maintained at Ghent University (Belgium) since 1996	University of Amsterdam	In-house culture, PERL	Aquatic Biosystems	Carolina Biological Supply	In-house culture, PERL	Oregon State University	Aquatica	Scientific Hatcheries
Test protocol	OECD 202 (OECD 1984b)	Modified OECD 202 (OECD 1984b)	General ASTM test guidelines; modified Gomot (1998)	USEPA (2002a)	ASTM International (2002a)	Newman (1975); Niederlehner et al. (1984)	ASTM International (2002a)	ASTM International (2002a)	ASTM International (2002a)	ASTM International (2002a)
Test duration	48 h	96 h	96 h	48 h	96 h	48 h	96 h	96 h	96 h	96 h
Culture and test medium	Artificial ISO medium (OECD 1998), 250 mg/L as CaCO ₃ , pH 7.5	Borgmann medium: 125 mg/L as CaCO ₃ , pH 7.9	Artificial freshwater: 140 mg/L as CaCO ₃ , pH 7.6–7.9	Synthetic moderately hard water, 100 mg/L as CaCO ₃	Lab-blended H ₂ O (well water blended with reverse osmosis H ₂ O)	Soft reconstituted lab water	Moderately hard reconstituted water	Lab-blended H ₂ O (well water blended with reverse osmosis H ₂ O)	Lab-blended H ₂ O (well water blended with reverse osmosis H ₂ O)	Lab-blended H ₂ O (well water blended with reverse osmosis H ₂ O)
Endpoints	Survival (immobilization)	Survival	Survival	Survival	Survival	Survival	Survival	Survival	Survival	Survival
Tested stage	<24 h old	6–7 d old	1 mo old	<24-h-old neonates	2nd instar larvae	<24 h old	Larval (3 d old)	Juvenile	Larval	Juvenile
Tested concentrations (nominal)	0.56–10 mg Co/L	0.56–18 mg Co/L	0–100 mg Co/L	0–8 mg Co/L	0–1000 mg Co/L	0–50 mg Co/L	0–10 mg Co/L	0–10 mg Co/L	0–40 mg Co/L	0–400 mg Co/L
Renewal frequency	Static	Static	Static	Static	Flow-through 14 mL/min	Static	Once at 48 h	Flow-through 10 mL/min	Flow-through 25 mL/min	Flow-through 50 mL/min
No. replicates/concentration	3	3	4	4	3	4	4	2	4	2

(Continued)

(Continued)

TABLE 2: (Continued)

	Cladoceran (<i>Daphnia magna</i>)	Amphipod (<i>Hyalella azteca</i>)	Great pond snail (<i>Lymnaea stagnalis</i>)	Water flea (<i>Ceriodaphnia dubia</i>)	Midge (<i>Chironomus tentans</i>)	Oligochaete (<i>Aeolosoma</i> sp.)	Fathead minnow (<i>Pimephales promelas</i>)	Rainbow trout (<i>Oncorhynchus mykiss</i>)	Zebrafish (<i>Danio rerio</i>)	
No. individuals/ replicate	10	10	5	5	10	5	10	20	10	
Test conditions	50-mL polyethylene cups, 20 °C, 16 L:8D, ~1000 lux	150-mL polyethylene cups, 100 mL/replicate, gauze substrate, 25 °C, 16 L:8D; ~1000 lux	500-mL glass jars, 400 mL/replicate, 20 °C, 12 L:12D, ~1000 lux	30-mL plastic cups, 25 mL/replicate, 20 °C, 16 L:8D	1-L glass jars, 200 mL/replicate, sterilized sand substrate, 23 °C, 16 L:8D	Polystyrene culture well plates, 10 mL/replicate, 25 °C, 16 L:8D	1-L beakers, 250 mL/replicate, 20 °C, 16 L:8D	19-L aquarium, 15 L/replicate, 12 °C, 16 L:8D	1-L chamber, 400–600 mL/replicate, 27 °C, 16 L:8D	19-L aquarium, 15 L/replicate, 25 °C, 16 L:8D
Feeding regime	None	None	None	None	Daily, Tetrafin slurry	None	Once, 2 h before 48-h renewal	None	Daily, infusoria and encapsulated shrimp diet	None

D = dark; ISO = International Organization for Standardization; L = light; OECD = Organisation for Economic Co-operation and Development; PERL = Parametrix Environmental Research Laboratory.

(milligrams per liter as CaCO_3), alkalinity (milligrams per liter as CaCO_3), total ammonia (milligrams per liter as N), total residual chlorine (milligrams per liter), dissolved oxygen (milligrams per liter), pH, and temperature (degrees Celsius) were measured in dilution water at test initiation and during the conduct of the tests. For the Pacific EcoRisk tests, water quality (e.g., temperature, pH, dissolved oxygen, conductivity, alkalinity, and total ammonia) was measured from one replicate per treatment. For the University of Ghent studies, pH, hardness, temperature, and oxygen concentration were measured at the beginning and/or end of each test (acute and chronic); and DOC was measured for all chronic tests.

In all of the Oregon State University studies both total and dissolved Co were measured in all treatments in each toxicity test, and in the University of Ghent tests only dissolved Co was measured. To quantify dissolved Co, treatment water was collected and passed through a 0.45- μm filter (Supor® polyethersulfone membrane; Acrodisc®; Pall Life Sciences) prior to acidification and analysis. Immediately after collection, total and filtered Co samples were acidified with nitric acid to a pH of <2 (0.14 N HNO_3 ; ultrapure). Cobalt concentrations were determined using flame or graphite furnace atomic absorption using a PerkinElmer AAnalyst™ 800 high-performance atomic absorption spectroscope according to USEPA methods 243.1 and 243.2 (US Environmental Protection Agency 1979; method detection level [MDL], 8 $\mu\text{g/L}$; limit of quantitation [LOQ], 25.3 $\mu\text{g/L}$; graphite furnace atomic absorption spectroscopy: MDL, 1.0 $\mu\text{g/L}$; LOQ, 3.1 $\mu\text{g/L}$). Dissolved Co was similarly analyzed for in the Ghent University studies.

The biological endpoint measured for all acute toxicity tests was survival (mortality was defined as no visible movement and no response to gentle prodding with a blunt probe). Survival was assessed at 48 h for *D. magna* (based on immobility), *H. azteca*, and *C. dubia* and 96 h for *C. tentans*, *L. stagnalis*, and the 3 fish species. Population growth (EC50) was considered as the acute toxicity endpoint assessed at 48 h for *Aeolosoma* sp. The chronic test biological endpoints included reproduction, growth, survival, and biomass. Reproduction and population growth were assessed for the rotifer *Brachionus calyciflorus* after 48 h and for *D. magna* after 21 d, growth rate was assessed for *H. azteca* and *L. stagnalis* after 28 d, and growth rate (biomass) was assessed for *P. subcapitata* after 72 h and for *L. minor* after 7 d. For *P. subcapitata*, exponentially dividing cells were filtered through a 0.45- μm membrane prior to analysis of the cell concentration. Survival and reproduction of *C. dubia* were assessed after 7 d of chronic exposure, population growth of *Aeolosoma* sp. was assessed following a 14-d chronic exposure, and survival and growth (biomass) were assessed for *C. tentans*, *P. promelas*, *O. mykiss*, and *D. rerio* following 20, 34, 81, and 33 d of chronic exposure, respectively. Reproductive endpoints for *C. tentans* were measured by inserting a removable emergence trap on day 20 to catch emerged adults. Emergence was recorded beginning on day 22, and adults were transferred to reproduction chambers. In the reproduction chambers, egg masses were examined daily, and the number of eggs was recorded.

Toxicity endpoints were assessed relative to the average dissolved Co concentrations. Acute LC50 values were

TABLE 3: Overview of the chronic test conditions with cobalt chloride for 9 different plant and invertebrate test species tested at Parametrix/Oregon State University and Ghent University (Chironomus tentans tests were conducted in an independent study by Pacific EcoRisk)

Origin	Cladoceran (<i>Daphnia magna</i>)	Amphipod (<i>Hyalalella azteca</i>)	Great pond snail (<i>Lymnaea stagnalis</i>)	Green alga (<i>Pseudokirchneriella subcapitata</i>)	Duckweed (<i>Lemna minor</i>)	Water flea (<i>Ceriodaphnia dubia</i>)	Oligochaete (<i>Aeolosoma</i> sp.)	Chironomus tentans	Rotifer (<i>Brachionus calyciflorus</i>)
Test protocol	OECD 211 (OECD 1998)	Modified OECD 211 (OECD 1998)	General ASTM test guidelines; modified Gomot (1998)	OECD 201 (OECD 1984a)	OECD 221 (OECD 2006)	USEPA (2000)	Newman (1975); Niederlehner et al. (1984)	USEPA (2000)	Snell and Moffat (1992), Grosell et al. (2006)
Test duration	21 d	28 d	28 d	72 h	7 d	7 d	14 d	20 d	48 h
Culture and test medium	Artificial ISO medium (OECD 1998), 250 mg/L as CaCO ₃ , pH 7.5	Borgmann medium (Borgmann 1996), 125 mg/L as CaCO ₃ , pH 7.9	Artificial freshwater, 140 mg/L as CaCO ₃ , pH 7.6–7.9	OECD medium (no EDTA; added DOM as Aldrich humic acid)	Modified Swedish standard medium (SSI 1995) (no EDTA; added DOM as Aldrich humic acid)	Synthetic moderately hard water, 108 mg/L as CaCO ₃	Synthetic soft water, 54 mg/L as CaCO ₃	Panther Creek water and lab control water at hardness matching that of Panther Creek	Moderately hard reconstituted water; hardness 100 mg/L as CaCO ₃
Endpoints	Survival and reproduction	Population growth	Growth rate	Biomass growth rate	Frond no., biomass (dry wt) of 2–4 visible fronds	Survival and reproduction neonates	Population growth	Survival, growth, reproduction <24 h old (1st instar larvae)	Population growth
Tested life stage	<24-h-old neonates	6–7 d old	1 mo old	1 × 10 ⁴ cells/mL of an exponential growing culture	Colonies of 2–4 visible fronds	<24-h-old neonates	<24 h old	<24 h old (1st instar larvae)	<2 h old
Tested concentrations (nominal)	5.6–320 µg/L Co	3.2–100 µg/L Co	3.2–1000 µg/L Co	18–560 µg/L Co	3.2–10 000 µg/L Co	0–200 µg/L Co	0–600 µg/L Co	0–2500 µg/L Co	0–2500 µg/L Co
Renewal frequency	Semistatic (renewal 3 times/wk)	Semistatic (renewal 3 times/wk)	Semistatic (renewal 2 times/wk)	Static; shaken 2 times daily	Static	Daily	3 times/wk	Flow-through, 0.6 mg/mL (2 times daily)	Static
No. replicates/concentration	10	10	8	3	4	10	4	12	4
No. individuals/replicate	1	3	1	1 × 10 ⁴ cells/mL of an exponential growing culture	12 fronds	1	Start of 5	12	25
Test conditions									

(Continued)

TABLE 3: (Continued)

Cladoceran (<i>Daphnia magna</i>)	Amphipod (<i>Hyalella azteca</i>)	Great pond snail (<i>Lymnaea stagnalis</i>)	Green alga (<i>Pseudokirchneriella subcapitata</i>)	Duckweed (<i>Lemna minor</i>)	Water flea (<i>Ceriodaphnia dubia</i>)	Oligochaete (<i>Aeolosoma</i> sp.)	Chironomus tentans	Rotifer (<i>Brachionus calyciflorus</i>)
50-mL polyethylene cups, 25 mL/replicate, 20 °C, 16 L:8D	100-mL polyethylene cups, 100 mL/replicate, 25 °C, 12 L:12D, ~1000 lux, gauze substrate	100-mL polyethylene cups, 100 mL/replicate, 25 °C, 12 L:12D, ~1000 lux, gauze substrate	100-mL Erlenmeyer, 50 mL/replicate, 25 °C, continuous light, 120 µmol photons m ⁻² s ⁻¹	500-mL glass jars, 100 mL/replicate, 24 °C, continuous light, 6000–10 000 lux	30-mL Souffle cups, 25 mL/replicate, 25 °C, 16 L:8D	Polystyrene culture well plates, 10 mL/replicate, 25 °C, 16 L:8D, 5 foot-candles	Flow-through setup in 300-mL glass beaker with 250-mL overflow, 23 °C, 16 L:8D	1-L chamber, 400–600 mL/replicate, 27 °C, 16 L:8D
Feeding regime Daily, 3:1 mix (<i>Pseudokirchneriella subcapitata</i> and <i>Chlamydomonas reinhardtii</i>)	Daily, 3:2 mix (YCT: <i>Pseudokirchneriella subcapitata</i>)	Lettuce (ad libitum)	None	None	Daily (algae/YCT)	Infusoria (rabbit pellet slurry)	1.5 mL slurried TetraMin flake fish food daily	Two times daily (brine shrimp)

D = dark; ISO = International Organization for Standardization; L = light; OECD = Organisation for Economic Co-operation and Development; PERL = Parametrix Environmental Research Laboratory; YCT = yeast–Cerophyl–trout chow.

TABLE 4: Overview of the chronic test conditions with cobalt chloride for 3 different fish test species tested at Parametrix/Oregon State University and Ghent University

	Fathead minnow (<i>Pimephales promelas</i>)	Rainbow trout (<i>Oncorhynchus mykiss</i>)	Zebrafish (<i>Danio rerio</i>)
Origin	In-house culture, PERL	Eggs and sperm in-house	Aquatica
Test protocol	ASTM International (2002b), OECD (1992)	ASTM International (2002b), OECD (1992)	ASTM International (2002b)
Test duration	34 d	81 d	33 d
Culture and test medium	Lab-blended H ₂ O (well water blended with reverse osmosis H ₂ O); hardness ~100 mg/L as CaCO ₃	Lab-blended H ₂ O (well water blended with reverse osmosis H ₂ O); hardness 115 mg/L as CaCO ₃	Lab-blended H ₂ O (well water blended with reverse osmosis H ₂ O); hardness 103 mg/L as CaCO ₃
Endpoints	Survival and growth	Survival and growth	Survival and growth
Tested life stage	Fertilized eggs	Fertilized eggs	Fertilized eggs (<48 h)
Tested concentrations (nominal)	0–3 mg/L Co	0–4 mg/L Co	0–8 mg/L Co
Renewal frequency	Flow-through, 25 mL/min	Flow-through, 22.5 mL/min	Flow-through, 7.5 mL/min
No. replicates/conc.	4	4	4
No. individuals/replicate	25	50 (start), thinned to 20	25
Test conditions	1-L chamber, 400–600 mL/replicate, 27 °C, 16 L:8D	1-L chamber, 600–800 mL/replicate, 12 °C, 16 L:8D	1-L chamber, 400–600 mL/replicate, 28 °C, 16 L:8D
Feeding regime	Two times daily (brine shrimp)	Two times daily (trout chow slurry)	Daily (infusoria and encapsulated shrimp diet)

D = dark; L = light; OECD = Organisation for Economic Co-operation and Development; PERL = Parametrix Environmental Research Laboratory.

calculated using a logistic equation and the Spearman-Kärber or trimmed Spearman-Kärber method, depending on the data collected for each test (Stephan 1977). Chronic statistical values were calculated using a logistic equation, threshold sigmoid regression, or piecewise linear regression analysis, based on best-fit analysis. Exposure concentrations were log(normal)-transformed before calculation of EC10 and EC20 values. All statistical calculations were carried out using the Comprehensive Environmental Toxicity Information System (Tidepool Scientific Software), the Toxicity Research Analysis Program (Ver 1.2; USEPA), or STATISTICA (StatSoft).

RESULTS

Acute toxicity tests

A total of 13 acute toxicity tests were conducted using 11 species. All water quality parameters (hardness, alkalinity, pH, conductivity, and temperature) remained within consistent tolerance ranges of the organisms throughout the present study and are presented in Table 5. Measured dissolved oxygen concentrations were >60% of saturation at the average test temperature. Actual concentrations of Co measured in the exposure solutions did not differ substantially from nominal exposure concentrations in static, static-renewal, or flow-through tests. In general, the dissolved Co concentrations for all studies were within 20% of the nominal values. Total Co concentrations averaged 76 to 123% of nominal Co concentrations, and the mean dissolved Co was approximately 82 to 120% of total. Comparisons of total and dissolved Co measurements indicated that the majority of the Co present in the exposure solutions was in a dissolved form. Control survival was >93% for all acute tests.

There was a significant concentration effect on juvenile and larval organism survival in all tests. Among the 11 species assessed, acute toxicity values (LC50) differed from a low of 90.1 µg Co/L for duckweed, *L. minor*, to a high of 429 000 µg Co/L for the midge, *C. tentans* (Table 5). The acute toxicity species sensitivity ranking, in order from most to least sensitive, was as follows: *L. minor* > *P. subcapitata* > *O. mykiss* > *C. dubia* > *P. promelas* > *H. azteca* > *D. magna* > *D. rerio* > *Aeolosoma* sp. > *L. stagnalis* > *C. tentans*. For the acute studies that included both larval and juvenile life stages of *P. promelas* and *D. rerio*, the larval fish were more sensitive by factors of 18 and 5, respectively, with larval and juvenile LC50 values of 3090 and 54 100 µg Co/L, respectively, for *P. promelas* and 15 980 and 85 290 µg Co/L, respectively, for *D. rerio* (Table 5). The LC50 values for all species used in acute Co bioassays are summarized in Table 5. All biological response data are presented in Supplemental Data, Tables S1 through S10.

Chronic toxicity tests

A total of 11 chronic toxicity studies were conducted using 11 species; one additional chronic test is discussed and included in the analysis that was conducted with the midge, *C. tentans*, and reported separately (Pacific EcoRisk 2005).

TABLE 5: Summary of acute toxicity test conditions and results for aquatic organisms exposed to cobalt chloride (micrograms of Co per liter dissolved)

Test species	Common name	Life stage	Hardness (mg/L as CaCO ₃)	Alkalinity (mg/L as CaCO ₃)	pH (SU)	DOC (mg/L)	E(L)C50 (95% CI)
<i>Lemna minor</i> ^a	Duckweed		55	6	6.5	0.3 ^b	90.1 (69.9–116.1)
<i>Pseudokirchneriella subcapitata</i> ^a	Algae		27	29	7.7	0.3 ^b	144 (118–176)
<i>Oncorhynchus mykiss</i>	Rainbow trout	Juvenile	60	60	7.2	0.5	1512 (1343–1704)
<i>Ceriodaphnia dubia</i>	Water flea	Neonates	100	100	8.1	<0.5	2154 (1566–2964)
<i>Pimephales promelas</i>	Fathead minnow	Larval	100	72	8.1	<0.05	3090 (2720–3520)
<i>Pimephales promelas</i>	Fathead minnow	Juvenile	118	112	7.7	0.6	54 100 (45 500–64 300)
<i>Hyalella azteca</i>	Amphipod	Juvenile	125	NR	7.9	NR	3290 (2920–3710)
<i>Daphnia magna</i>	Water flea	Juvenile	250	NR	7.8	NR	5890 (5680–6100)
<i>Danio rerio</i>	Zebrafish	Larval	100	108	7.7	0.5	15 980 (13 630–18 730)
<i>Danio rerio</i>	Zebrafish	Juvenile	120	92	7.8	0.6	85 290 (72 300–100 700)
<i>Aeolosoma</i> sp.	Worm	Neonate	48	80	7.7	<0.05	42 700 (39 680–45 960)
<i>Lymnaea stagnalis</i>	Snail	1 mo old	140	NR	7.2–7.7	NR	61 600 (44 100–86 100)
<i>Chironomus tentans</i>	Midge	2nd-instar larvae	84	108	7.6	0.5	429 000 (359 000–512 700)

^aNot included in calculation of an USEPA AWQC final acute value.

^bEstimate of "initial" DOC of the dilution medium (based on earlier measurements of laboratory deionized water); does not account for DOC development/excretions during tests. E(L)C50 = median effect or lethal concentration; NR = not reported.

A summary of water quality characteristics for the control/dilution waters in all of the chronic tests is provided in Table 6. Measured concentrations of Co in the exposure solutions did not vary substantially over the test duration and did not differ substantially from nominal exposure concentrations in either the static-renewal or flow-through tests. The time-weighted averages over the duration of the tests were between 85 and 126%. Total Co concentrations averaged 105 to 119% of nominal Co concentrations, and the mean dissolved Co was approximately 89 to 105% of total. Comparisons of total and dissolved Co measurements indicated that the majority of the Co present in the exposure solutions was in a dissolved form; thus, all data endpoints are reported on the basis of dissolved Co. Chronic effect levels for 12 species tested were between 4.9 µg Co/L (EC10 for *L. minor*) and 2171 µg Co/L (EC10 for *O. mykiss*; Table 7). These values differed by a factor of 443, underscoring the wide range of sensitivities among the species tested. The chronic toxicity species sensitivity ranking, in order from most to least sensitive, was as follows: *L. minor* > *H. azteca* > *C. dubia* > *L. stagnalis* > *P. subcapitata* > *D. magna* > *Aeolosoma* sp. > *C. tentans* > *P. promelas* > *B. calyciflorus* > *D. rerio* > *O. mykiss*. The EC10 and EC20 values following chronic Co exposure are summarized in Table 7. Acute-to-chronic ratios, defined as the acute LC50 divided by the chronic EC20, differed substantially, ranging from 0.6 (*O. mykiss*) to 2670 (*L. stagnalis*), with invertebrate species showing larger acute-to-chronic ratios (between 129 and 2670, $n = 6$) than those noted for fish species (between 0.6 and 10, $n = 3$; Figure 1). The acute-to-chronic ratios defined as the acute LC50 divided by the chronic EC10 were similar to the acute-to-chronic ratios based on EC20 and ranged from 0.7 (*O. mykiss*) to 6410 (*L. stagnalis*), with invertebrate species showing larger acute-to-chronic ratios (between 182 and 6410, $n = 6$) than those noted for fish species (between 0.7 and 14.7, $n = 3$; Figure 1).

In addition to the “most sensitive” endpoints summarized in Table 7 (e.g., growth, biomass), mean time-to-hatch (days) and presence or absence of morphological defects (i.e., teratogenicity) were recorded, if observed, for 3 of the fish species. For *P. promelas*, hatching success was not affected by Co exposure over the series of tested concentrations. The overall median time-to-hatch for 5 treatments was 3.94 ± 0.4 d compared to 3.5 ± 0.6 d in controls. For *D. rerio*, the overall median time-to-hatch for 5 treatments was 4.58 ± 0.6 d compared to 4.3 ± 0.5 d in controls; no treatment-related differences in hatch time were noted. For *O. mykiss*, no differences in egg mortality or time-to-hatch were observed. The overall median time-to-hatch for 5 treatments was 34.7 ± 0.26 d, in comparison to 34.0 ± 0.2 d in the controls. No malformations were observed in any of the controls or Co-exposed organisms during any of these studies. All biological response data are presented in Supplemental Data, Tables S11 through S22.

DISCUSSION

Evaluation of generated data

To our knowledge, the present study represents the first comprehensive report of Co's acute and chronic freshwater

TABLE 6: Water quality results from chronic toxicity tests

Parameters	<i>Pseudokirchneriella subcapitata</i>	<i>Lemna minor</i>	<i>Ceriodaphnia dubia</i>	<i>Daphnia magna</i>	<i>Hyalella azteca</i>	<i>Chironomus tentans</i>	<i>Aeolosoma</i> sp.	<i>Brachionus calyciflorus</i>	<i>Lymnaea stagnalis</i>	<i>Pimephales promelas</i>	<i>Danio rerio</i>	<i>Oncorhynchus mykiss</i>
Average hardness (mg/L as CaCO ₃):	27.4	54.9	46.7	250.3	125.2	34.3	50.1	52	140.1	98.4	98.4	115.8
Average alkalinity (mg/L as CaCO ₃):	28.6	5.8	76	36.3	46.8	34	42	72	116.2	120	117	130
Average pH in chamber (SU):	7.7	6.5	8.3	7.5	7.5	8.2	7.6	7.4	7.8	8.0	7.8	7.8
DOC (mg/L):	0.3	0.3	0.5	0.3	0.3	2.2	0.5	0.4	0.3	0.5	0.5	1.1
Calcium (mg/L):	4.91	9.81	8.2	80.1	40.1	8.8	12.8	11.5	40.1	22.3	22.3	26.8
Magnesium (mg/L):	3.7	7.4	6.4	12.2	6.08	3.0	4.41	5.32	9.73	10.4	10.4	11.9
Sodium (mg/L):	15	33.7	23.7	17.7	23.2	15	16.1	14.2	55.1	31.3	31.3	42.2
Potassium (mg/L):	0.46	3.74	2.2	3.04	1.96	1	1.6	0.382	11.7	1.14	1.14	0.8
Chloride (mg/L):	26.7	17.7	13.5	144.6	72.7	1	1.4	12.9	99.3	21.1	43.2	47.8
Sulfate (mg/L):	5.9	29.2	61.7	48.1	24.0	15	46.0	0.6	14.4	0.3	0.3	0.6

TABLE 7: Summary of chronic toxicity data for aquatic organisms exposed to cobalt chloride (micrograms of Co per liter dissolved)

Test species	Common name	Most sensitive endpoint	EC10 (95% CI)	EC20 (95% CI)
<i>Lemna minor</i>	Duckweed	7-d growth rate	4.9 (2.7–8.7)	NC
<i>Hyalella azteca</i>	Amphipod	Growth rate (28-d dry weight)	7.55 (4.00–14.27)	17.58 (11.92–14.27)
<i>Ceriodaphnia dubia</i>	Water flea	7-d reproduction	7.89 (0.72–86.37)	11.08 (1.88–65.29)
<i>Lymnaea stagnalis</i>	Snail	28-d growth rate	9.61 (3.65–25.24)	23.07 (12.03–44.22)
<i>Pseudokirchneriella subcapitata</i>	Algae	72-h growth rate	23.0 (14.1–37.5)	NC
<i>Daphnia magna</i>	Water flea	21-d net reproduction (no. young/female)	32.4 (21.8–48.0)	45.7 (34.5–60.4)
<i>Aelosoma</i> sp.	Worm	14-d growth	154.6 (124.9–191.5)	185.4 (154.3–222.7)
<i>Chironomus tentans</i>	Midge	20-d survival	167.1 (104.8–266.4)	225.2 (150.1–337.9)
<i>Pimephales promelas</i>	Fathead minnow	34-d survival	351.4 (210.6–586.5)	409.0 (268.1–623.9)
<i>Brachionus calyciflorus</i>	Rotifer	Population, no. of individuals	442.2 (221–887)	784.9 (478–1289)
<i>Danio rerio</i>	Zebrafish	33-d biomass	1085 (569–2068)	1593 (946–2682)
<i>Oncorhynchus mykiss</i>	Rainbow trout	81-d biomass	2171 (1658–2842)	2495 (1995–3120)

EC10/EC20 = 10 and 20% effect concentrations, respectively; NC = not calculated.

toxicity to species representing taxa ranging from algae/plants to invertebrates to fish. Among the organisms tested, *L. minor* showed the greatest sensitivity to Co under acute and chronic exposure. The acute species sensitivity ranking demonstrated that juvenile rainbow trout was the most sensitive nonplant species tested, with an LC50 of 1512 (1343–1704) $\mu\text{g Co/L}$. Interestingly, this species was the least sensitive in chronic toxicity testing. However, it is important to note that the acute test was performed with juvenile rainbow trout, whereas the chronic test was initiated with newly fertilized eggs. Differences in life-stage sensitivities likely contributed to this discrepancy. De Schampelaere and Janssen (2004) reported, similarly, that chronic exposures of fish eggs to zinc resulted in lower toxicity than exposures starting with juvenile fish.

Prior to the studies described herein, little reliable chronic toxicity data existed for Co. Biesinger and Christensen (1972) used CoCl_2 as the Co test compound and reported a *D. magna*

21-d EC16 (reproduction) value of 10 $\mu\text{g Co/L}$ (nominal) when tested in Lake Superior water (hardness 45 mg/L as CaCO_3 , pH 7.7). In the present study, we found the 28-d *D. magna* EC20 to be 65.2 $\mu\text{g Co/L}$, which is approximately 5-fold higher than the EC16 value reported by Biesinger and Christensen (1972). As a direct comparison between the studies, Biesinger and Christensen (1972) reported an acute (48-h) LC50 of 1.11 mg Co/L, whereas the present study reported a 48-h LC50 of 5.89 mg Co/L. This reflects an approximate 5-fold difference in acute Co sensitivity between the 2 studies, which is consistent with the apparent difference in chronic Co sensitivity. This discrepancy could be attributable to disparities in the *D. magna* clones used previously compared with those used in the present study or other methodological inconsistencies (e.g., one time vs 3 times weekly solution renewals), differences in the statistical endpoints (EC16 vs EC20), and possible exposure concentration differences (Biesinger and Christensen [1972] reported data based on nominal concentrations).

Aside from the Biesinger and Christensen (1972) study, other studies published on Co freshwater toxicity differed from the present study in that they focused on acute exposures for which the duration differed from that in the present study. For example, Diamond et al. (1992) reported 48-h no-observed-effect concentrations (NOECs) for *P. promelas* of 1.25 (hardness 50 mg/L as CaCO_3) to 13.7 mg Co/L (hardness 400 mg/L) and 7-d NOECs of 1.23 (hardness 50 mg/L) to 3.83 mg Co/L (hardness 800 mg/L). Diamond et al. (1992) also reported 24-h LC50 values of 2.35 (hardness 50 mg/L) to 4.20 mg Co/L (hardness 400 mg/L) for *C. dubia*, which is similar to the 48-h LC50 of 2.39 mg Co/L (moderately hard water, hardness 100 mg/L) reported in the present study. Marr et al. (1998) reported a 96-h LC50 of 1.41 mg Co/L for *O. mykiss*, which was consistent with the 96-h LC50 of 1.51 mg Co/L reported in the present study (Table 4). The results of the Diamond et al. (1992) study underscore the importance of considering water hardness because toxicity appears to be a function of this factor. Notably, the present study did not consider bioavailability; however, this is being addressed in subsequent research. Thus, although limited chronic toxicity data are available for comparison (e.g., Norwood et al. 2007; G.L. Kimball, University of Minnesota, Minneapolis, MN, unpublished manuscript), acute toxicity

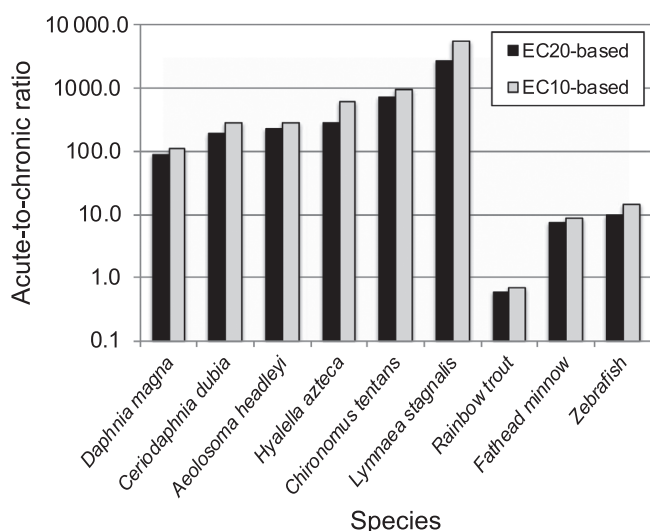


FIGURE 1: Effect concentrations, 10 and 20%, for 9 freshwater species exposed to CoCl_2 . Data presented in order of relative chronic Co sensitivity (most sensitive on left, least sensitive on right). Fish species have lowest chronic Co sensitivity and lowest acute-to-chronic ratios. EC10/EC20 = 10 and 20% effect concentrations, respectively.

results were generally consistent with those studies previously reported in the literature. Chronic toxicity studies that were identified typically did not meet current scientific standards for acceptability (e.g., lack of measured concentrations, unavailability of raw data with which to do statistical analyses) and were therefore not included in our SSD calculations.

Derivation of an SSD

The primary objective of the present study was to generate data that could be used to derive a freshwater PNEC ($\text{PNEC}_{\text{freshwater}}$) in accordance with European criteria (European Chemicals Agency 2008) and an AWQC-style aquatic reference standard following USEPA methods (US Environmental Protection Agency 1985). The $\text{PNEC}_{\text{freshwater}}$ can be derived using a limited database in conjunction with specific assessment factors (10–100) or through the use of an SSD analysis on a more “robust” data set (assessment factor 1–5). A minimum of 8 different taxonomic groups should be included in the effects database for SSD analysis, and this guidance was adopted by the ECHA (European Chemicals Agency 2008; Table 1). The Co chronic effects studies described herein used organisms representing each of these categories, including a range of algal, invertebrate, fish, and plant species (Table 1).

The 12 chronic freshwater toxicity EC10 values generated for 12 different freshwater organisms (Table 5) were used for the construction of an SSD ($n = 12$ species; Figure 2). Using the software package ETX (Ver 2.0), a log-normal distribution was fitted to the data set and the hazardous concentration for 5% of the species (HC5 value) was derived as the 5th percentile of the SSD using a log-normal distribution function (Van Vlaardingen et al. 2004). According to the ECHA guidelines, we derived the median estimate of the HC5 value, the $\text{HC}_{5,50\%}$, which was $1.80 \mu\text{g Co/L}$ (95% CI $0.16\text{--}7.62 \mu\text{g Co/L}$; Figure 2). The “fit” of the log-normal distribution function is not ideal, and an analysis of the “best fit” for available functions (@RISK software; Palisade Corporation) indicates that a Pareto distribution fits the data more appropriately, resulting in an HC5 of $5.62 \mu\text{g Co/L}$.

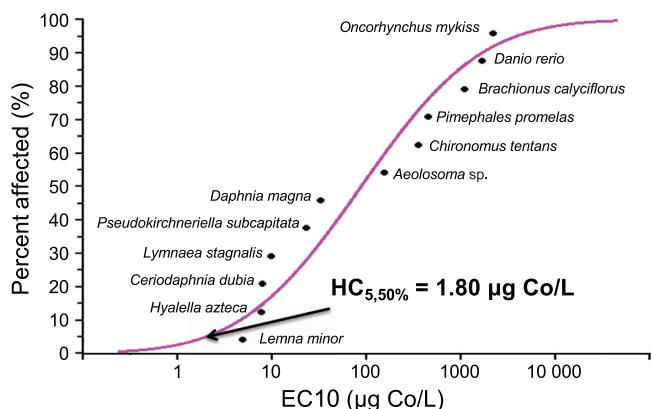


FIGURE 2: Species sensitivity distribution and $\text{HC}_{5,50\%}$ fit to chronic EC10 values for Co in aquatic environments. EC10 = 10% effect concentration; $\text{HC}_{5,50\%}$ = median hazardous concentration for 5% of the organisms.

A variety of statistical methods exist for determining the acceptability of “goodness-of-fit” for a particular distribution model. The Akaike information criterion, the Bayesian information criterion, the Anderson-Darling goodness-of-fit test, and the Kolmogorov-Smirnov test have been suggested as criteria to aid in the choice of a parametric distribution (Wang and Liu 2006; European Chemicals Agency 2008; International Council on Mining and Metals 2016). All of these assessment metrics were conducted and showed that the Pareto distribution provided a superior fit for the SSD data than the logistic model.

According to the Registration, Evaluation, Authorisation, and Restriction of Chemicals (REACH) regulations (European Chemicals Agency 2008), if the SSD technique is used, an assessment factor between 1 and 5, to be judged on a case-by-case basis, should be applied to the HC5 (i.e., $\text{PNEC}_{\text{aquatic}} = \text{HC}_{5,50\%}/\text{assessment factor}$). The size of the assessment factor should reflect the degree of uncertainty associated with the robustness and scope of the available data. Selection of an assessment factor should be based on an uncertainty analysis best suited to risk assessors and is therefore not within the scope of this article.

The present study also provided an acceptable database for use in deriving both an acute (CMC) and a chronic (CCC) AWQC-style aquatic reference standard following the USEPA-prescribed approach (US Environmental Protection Agency 1985). Using the acute toxicity data presented in Table 4 (not considering the plant and algal data per the USEPA methodology), the final acute value (FAV) would be $1199 \mu\text{g Co/L}$; dividing this value by 2 would give a CMC of $599.5 \mu\text{g Co/L}$. When sufficient data are available to characterize the effect of a water quality parameter (e.g., water hardness, pH, DOC), this information can be used to normalize the available data to a common condition (e.g., a common water hardness), thereby reducing between-test variability within a test species. Insufficient acute toxicity data are currently available to permit this type of correction with Co, although relationships between water hardness and Co acute toxicity have been reported (Diamond et al. 1992; Pourkhabbaz et al. 2011; Blust 2012; Environment Canada 2017).

Although use of an SSD-based approach is favored in deriving chronic AWQC, limitations in the availability of chronic toxicity data frequently lead to the derivation of AWQC CCC by dividing the FAV by the acute-to-chronic ratio. However, the range of acute-to-chronic ratios observed with Co is too large to apply this method for deriving the CCC (US Environmental Protection Agency 1985); therefore, the SSD approach based on chronic data must be employed. Based on the studies reported herein, sufficient chronic data are available to characterize the effect of water quality parameters on Co chronic toxicity. Using this information and following the USEPA's method for AWQC derivation, the USEPA-style final chronic value (FCV) is $7.13 \mu\text{g Co/L}$. This is similar to the PNEC derived in the present study of $5.62 \mu\text{g Co/L}$ (using a “best-fit” procedure). Thus, despite differences in approaches to determining PNECs versus an AWQC-style aquatic reference standard, the outcome was similar.

As noted in Table 1, there are differences in the requirements set forth for Europe (under REACH) versus the USEPA (pursuant to Section 304[a] of the Clean Water Act). For example, whereas REACH requirements do not consider acute toxicity, much of the data used for establishment of AWQC for most of the metals in the United States were derived from acute toxicity data in cases where chronic toxicity data were not available and derivation of chronic AWQC were based on an acute-to-chronic ratio approach (US Environmental Protection Agency 1985). In addition, the United States does not include plant or algal data in developing the SSD used to derive the FCV subsequently used in establishing the AWQC (Table 1). Ideally, both acute and chronic freshwater toxicity data should be considered in deriving aquatic standards. As we demonstrated in the present study, generation of acute toxicity data is an important prelude to chronic toxicity testing. Careful planning so that both acute and chronic toxicity testing follows the requirements of the respective agencies will provide the most useful toxicity information to an international audience.

CONCLUSION

Due to the lack of published data on Co freshwater toxicity, a thorough assessment of acute and chronic toxicity was warranted. We conducted a series of freshwater acute and chronic toxicity assays using 12 species (including one from an external report), ranging from algae to fish and including the 8 taxonomic groups required by the ECHA for derivation of a PNEC and the USEPA for derivation of an AWQC. All other REACH requirements for chronic toxicity testing were fulfilled, in addition to fulfilling AWQC requirements established by the Clean Water Act in the United States, making the data internationally relevant (US Environmental Protection Agency 1985). Chronic toxicity EC10 values ranged from 4.9 $\mu\text{g Co/L}$ (*L. minor*) to 2171 $\mu\text{g Co/L}$ (*O. mykiss*), representing a 443-fold difference. An SSD was used to calculate an HC5,50% of 1.80 $\mu\text{g Co/L}$ (or 5.62 $\mu\text{g Co/L}$ using a “best-fit” analysis). The existing literature and these data highlight the need for developing both acute and chronic toxicity assessments. Owing to the effect of site-specific water quality parameters (e.g., hardness, pH) on the bioavailability and toxicity of metals, future assessments of Co toxicity will build on the data reported herein and will address the effects of these parameters.

Supplemental Data—The Supplemental Data are available on the Wiley Online Library at DOI: 10.1002/etc.4662.

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Data Availability Statement—Data, associated metadata, and calculation tools are available from the corresponding author (bill.stubblefield@oregonstate.edu).

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